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MIL-C-70567B (AR)
02 May 1994
SUPERSEDING
MIL-C-70567A (AR)
23 Februa~ 1987

MILITARY SPECIFICATION

CIRCUIT BOARD, COMPUTER AND MEMORY: 9360878

This specification is approved for use by the U.S. ArmY Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the fire control processor computer and memory circuit card assembly, referred to as the computer and memory board (CmB). The CMB is used to compute ballistics calculations and communicate with the semo control processor input/-output board and data transmission beard. The CMB also manages weapon syst~ Built-In-Test (BIT). The CMB is a 10 1/2 inch by 7 1/4 inch double-layered, multi-wire circuit card mounted vertically in the fire control processor (FCP). Refer to Figure 1 for the CMB functional block diagram (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Goverment documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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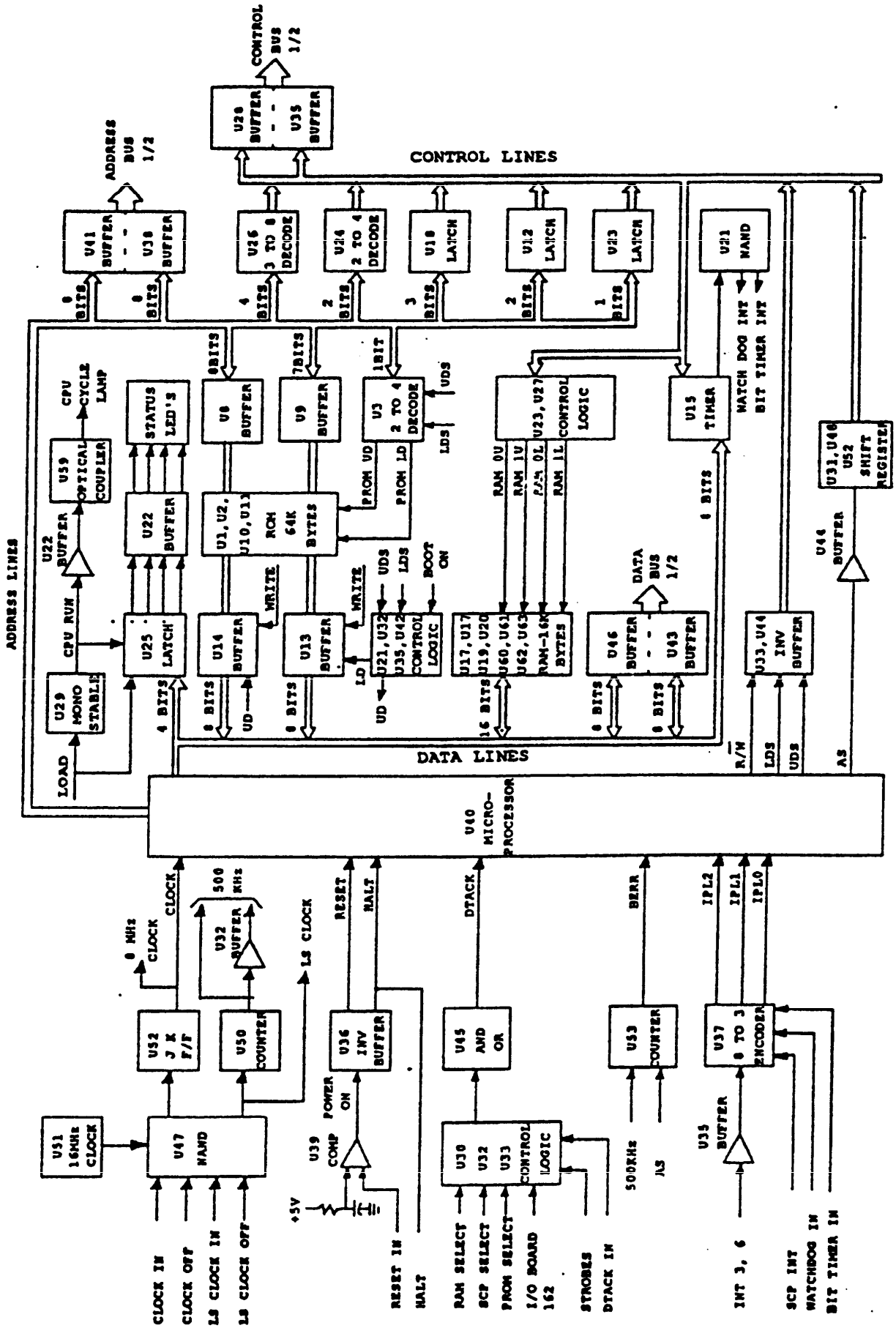


FIGURE 1. CMB functional block diagram.

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SPECIFICATIONS

MILITARY

MIL-F-13926	**Fire Control Materiel, Manufacture and Inspection, General Specification for**
MIL-I-45607	**Inspection Equipment,** ***Acquisition,*** **Maintenance and Disposition of**

STANDARDS

MILITARY

MIL-STD-109	**Quality Assurance Terms** and Definitions
MIL-STD-810	Environmental Test Methods and Engineering Guidelines

unless **otherwise indicated, copies of federal and military specifications, standards~ and handbooks are available from the DODSSP - Customer Service, Standardization** Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360878	-	**Computer and Memory** Assembly
9360880	-	Computer and Memory Schematic
9361334	-	Test Program Set **Manual**
9361380	**-**	**Interface Device Adopter Board A**
9397497	**-**	**Cable** Assembly W1
9397498	-	Test Plug P3
9397499	-	Test Plug P5
12563341	-	Software Control Drawing
12950877	-	Requirements for Soldering Electrical and Electronic Assemblies

PACKAGING ***DATA*** **SHEET**

PS 9360878	**-**	**Packaging of Computer** and Memory Assembly

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(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government Publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ANSI/IPC-DW-425 - Design and End Product Requirements for Discrete Wiring Boards

(Application for copies should be addressed to the American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

2.3 order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360878, all associated drawings, and with all requirements specified in applicable specifications and drawings. Where applicable, the CMB shall meet the requirements of MIL-F-13926 and ANSI/IPC-DW-425 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

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3.2.2 ANSI/IPC-DW-425.

a. IPC-DW-425/11 me Cl plated through connection.

3.3 Erwiromental service conditions.

3.3.1 Vibration. The CMB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to **the** equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The CMB shall not suffer damage and shall meet the requirements *of 3.4* and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10 percent, 11 milliseconds ± 10 percent, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The CMB shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level **(20.6** inches of **Hg).**

3.3.4 Temperature.

3.3.4.1 operating. The CMB shall meet the requirements of 3.4 and 3.S when stabilized thermally at any temperature between -40"F and +155°F.

3.3 .4 .2 Non-operating (storage). The CMB shall not suffer **permanen**t damage when subjected to the constant temperature **extremes** of -60"F for a period of 24 hours and +160°F for a **period o**f 4 hours (after temperature stabilization), and sha11 **meet the** requirements of 3.4 and 3.5 when returned to room **ambient.**

3.3.5 Hu midity (non-o~erat ins) The CMB shall meet the requirements ***of*** *3.4* and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 **days.**

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagr- 9360880 of the cm.

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3.5 Performance.

3.5.1 Functional description. The CMB shall contain the main central processing unit (CPU) and memory devices which control the VADS fire control processor. The CPU is a 68000 microprocessor which runs at an 8 MHz rate. The data and address buses are 16 and 21 bits respectively. The memory shall consist of 128K bytes of ROM and 8K bytes of RAM expandable to 16K bytes of RAM. The primary function shall be the calculation of ballistics solutions and direction and control of the associated interfaces and servos.

3.5.2 Power requirements. The CMB power requirements shall be as specified in Table I.

TABLE **1. Power requirments.**

Parameter	+ 5 Vdc	+ 24 Vdc
Voltage Tolerance	A 0.25 Vdc	* 2.4 Vdc
Maximum Current	3 A	0.1 A
Maximum Ripple	0.1 V Peak-to-Peak	0.5 V Peak-to-Peak
Max. Source Impedance	0.033 ohms	0.048 *OhIls*
Input Terminals	PI-32 thru 34, 72 thru 74 EPROM VCC: P2-32 EPROM VPP: P2-33 NOTE &/	NOTE ~/
Input Return	PI-35 thru 37, 75 thru 77, 9, 49, 18, 58	P2-34

Notes :

1/ VPP (P2-33) may be obtained by connecting to vcC (P2-32), except during programming.

2/ The +24 vdc shall be applied through a 24 V, 40 mA lamp (or equivalent resistance) to P2-35, 70

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3.5.3 clock circuit. With power per 3.5.2 applied to the CMB, the **16 MHZ output of the on-board clock oscillator shall be** divided by two to form an 8 MHz ± 1,176 Hz TTL square wave clock signal at CLOCK, J1-38 and LS CLOCK (8 MHz)t p170. **Rise and fall times for the 8 MHz clock outputs shall be 10 nanoseconds maximum each. Inputting a logic low on INTERNAL CLOCK DIs-LE# J1-2, shall disable the on-board 8 MHz clock outputs and allow for an external clock to be input to the CMB at EXT CLOCK, J1-27, for test purposes. The 8 MHz clock shall be divided by 16 to form a 500 KHz * 74 Hz TTL square wave clock signal at CLOCK (500 KHz), P1-50. Inputting a logic low on LS CLOCK OFF, J1-1, shall disable the 500 KHZ clock output and allow for an external low speed test clock to be input at LS CLOCK IN, J1-26.**

3.5.4 Power-on and manual reset circuit.

3.5.4.1 Power-on circuit. When power per 3.5.2 is initially applied to the CMB, POWER ON (Pi-69), TEST RESET (J1-9), RESET (Pi-29) and HALT (J1-38) shall remain logic low for 150 * 50 milliseconds, causing a logic low on the MC68000 microprocessor's RESET (u40-18) and HALT (u40-17) inputs. The MC68000 address and data bus lines shall be in a high-impedance state and render its control signals inactive per Table II. After the power-on control lines shall become active and the start-up BIT routine in EPROMS U1O and Ull shall be initiated.

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TABLE xx. *MC68000* reset condition.

Date Lines in High-Impedance State	**Address Lines in** High-Impedance State	Control Lines Inactive
D0 (J1-44) D1 (J1-19) D2 (J1-20) D3 (J1-45) D4 (J1-5) D5 (J1-28) D6 (J1-4) D7 (J1-29) D8 (J1-21) D9 (J1-46) D10 (J1-22) Dll (J1-47) D12 (J1-30) D13 (J1-6) D14 (J1-31) D15 (J1-7)	Al (P2-17) A2 (P2-18) A3 (P2-55) A4 (P2-20) AS (P2-19) A6 (P2-57) A7 (P2-59) A8 (P2-24) A9 (P2-58) A10 (P2-23) All (P2-60) A12 (P2-25) A13 (P2-26) A14 (P2-28) A15 (P2-62) A16 (P2-27) A17 (P2-37) A18 (P2-2) A19 (P2-36) A20 (P2-1)	AS (J1-32) high WRITE (J1-34) high LDS (J1-33) high UDS (J1-8) high FC2 (J1-16) low FC1 (J1-41) low FCO (J1-17) 10W

3.5.4.2 Start -up BIT. The start-up BIT routine shall run for less than 3 seconds. Start-up BIT shall test EPROMS Ul, U2, u10, u1l, RAM **integrated circuits U16, U17, U19, U20 and the Arithmetic Logic Unit (ALU) circuit within microprocessor U40. The start-up BIT routine shall:**

a. Conduct a checksum of each EPROM and compare it to a stored value in EPROM to verify program.

b. Write and read AAAAH (H means hexadecimal) then 5555H to RAM to ensure all bits toggle.

c. Write each RAM address number into its address and read it back, checking it for proper addressing.

d. Perform mathematical operations with the ALU to determine if it is functioning properly.

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The start-up BIT routine shall set (logic high) flags at the addresses of Table III for each item tested successfully; otherwise, the flag bit shall be cleared (logzc low) bits of Table III shall be latched logic high following the start-up BIT test.

TABLE XII. start-up BIT test flags

Address	Flag Bit	Item Tested
60022H	DO	EPROM U11
60022H	D1	EPROM U1O
GO022H	D2	EPROM Ul
60022H	D3	EPROM UZ
GO022H	D8	u u16
GO022H	D9	w U20
60022H	D1 0	w U17
60022H	Dll	w U19
60011H	D()	ALu U40

Start-up BIT results shall be latched and displayed On LEDS DS4 through DS1 per Table XV, showing the CMB tested 'good communications with other system boards was not possible.

TABLE IV. Start-un BIT LED status.

LED	Test Result	Circuit Board Represented
DS4	Flashing at 4 Hz	Display Control Panel (DCP)
DS3	Flashing at 4 Hz	Input/Output Board (IOB)
DS2	Flashing at 4 Hz	Servo Controller Board (SCB)
DS1	Lit	Computer and Memom Board (-)

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3.5.4.3 With power applied to the CMB per 3.5.2, bringing TEST RESET (J1-9) logic low for at least 10 microseconds, RESET (PI-69), RESET (Pi-29), and HALT (J1-38) to go logic low, the microprocessor's address and data bus lines shall go to a high-impedance state and causing the microprocessor's control lines to deactivate per 3.s.4.1. When TEST RESET is allowed to return high, start-up BIT per 3.5.4.2 shall run.

3.5.4.4 With power applied to the CMB per 3.5.2, bringing HALT (J1-38) iogic low shall cause the microprocessor to stop executing instructions at the end of the current bus cycle, set its address and data bus lines to a high-impedance state and render its control lines inactive per 3.5.4.1. Bringing HALT logic high shall cause the microprocessor to continue execution with the next bus cycle.

3.5.5 MC60000 microprocessor u40.

3.5.5.1 control signals The Main MC68000 control signals shall be address strobe AS (Ji-32), WRITE (J1-34), lower data strobe LDS (J1-3~per data strobe UDS (Jlk&L, data transfer acknowledge DTACK (J1-10) and bus error BERR (J1-14). During every bus cycle, the MC68000 shall output the address bits onto address lines Al through A20, then assert AS logic low kQ indicate there is a valid address on the address bus. A DTACK signal shall normally initiate termination of the bus cycle. If the DTACK signal is not received within 16 microseconds following AS going logic low, BERR shall go logic low to notify the MC68000 that a problem exists with the cycle under execution, therefore, terminate the cycle and setting the address and data bus lines to a high-impedance state. with BERR low, AS shall be released and go high, AS shall go low, presetting countdown counter US3, and BERR shall go high. With BERR high, the MC68000 shall again attempt to rerun the same bus cycle by loading the same address on the address bus and asserting AS low {AS high). If a DTACg signal is not received within 16 microseconds a second time, BERR shall again go logic low and halt microprocessor operation. The address and data buses shall be set to a high-impedance state and the control lines shall become Inactive. The MC68000 shall only restart if it receives a logic low on TEST RESET (J1-9) for more than 1.4 microseconds or if CMB power is removed then reapplied to operate the power-on reset circuit.

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3.5.5.2 MC68000 **read d write cycle operation.**

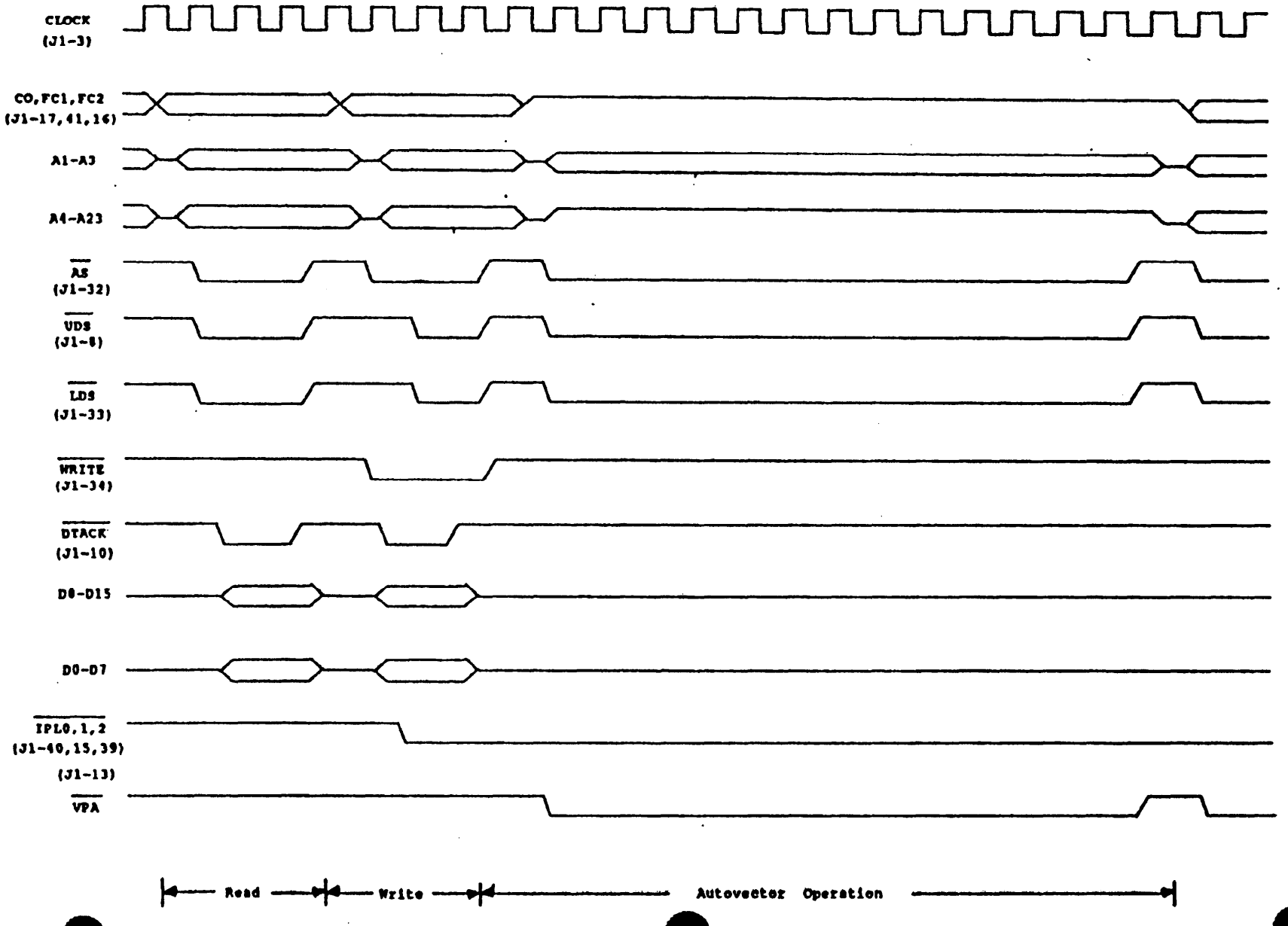
3.5.5.2.1 **ad IICvcu During a single read bus cycle, microprocessor u40 shall received data from memory or a peripheral device and read it by bytes per the following sequence of events illustrated in Figure 2.**

a. The MC68000 shall:

1. **set WRITE (J1-34) logic high to perform a read operation (READ, Pi-24, shall be logic low**
2. **Set function code outputs FCO, FCl and FC2 per Table V for the cycle type being executed.**
3. **place an address on address lines Al through A23.**
4. **Assert AS (J1-32) logic low.**
5. **Assert UDS (J1-8), (J1-33)** or both logic low, depending on whether the software instruction being executed requests an upper byte, lower byte or word read. If a byte read is requested, the internal AO bit is decoded as follows:

   For AO logic low, UDS is asserted.
   For AO logic high, LDS is asserted.

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TABLE v. **Function code output.**

Function Code Output			
FC2(J1-16)	FC1(J1-41)	FCO(J1-17)	Cycle me
L	L	L	(Undefined, Reserved)
L	L	H	User Data
L	H	L	User Program
L	H	H	(Undefined, Reserved)
H	L	L	(Undefined, Reserved)
H	L	H	Supewisor Data
H	H	L	Supervisor Progr~
H	H	H	Interrupt Acknowledge

b. **The memory or peripheral device addressed shall:**

1. **Decode the address on the address bus lines Al through A20.**

2. **Place data on data lines DO through D7 and D8 through D15 as required.**

c. **The** CMB DTACK **circuit shall automatically assert DTACK logic low after a set time delay following AS going low.**

d. The MC68000 shall **upon receipt of DTACK low:**

1. Latch the data **on the data bus.**

2. Negate UDS and **LDS (as asserted) .**

3. Negate AS.

e. The memory or peripheral device addressed shall remove the data **from** DO through D7 and D8 through D15 (as **asserted) .**

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3.5.5.2.2 Write Cycle. During a single write bus cycle, microprocessor u40 shall send data to a memory or Peripheral device and write it by bytes per the following sequence of events; illustrated in Figure 2.

. a. The MC68000 shall:

1 0 Set function code outputs FCO, FC1 and FC2 per Table V.

2. Place an address on address lines Al through A23.

3. Assert = (J1-32) logic low.

4. Set $\overline{\text{WRITE}}$ (J1-34) loglc low to perform a write operation (READ, Pi-24, shall be 109IC high.

5. place data on data lines DO through D7 and D8 through D15, depending on the AO bit as follows for a byte write:

For *AO logic* low, D8 through D15.
For AO logic high, DO through D7.
For a word write, data shall be placed on DO through D15 per the software instruction.

60 Assert UDS (J1-8) *or* LDS (J1-33) logic low depending on the AO bit *as* follows:

For AO logic low, UDS shall be asserted for an upper byte write.

For AO logic high, LDS shall be asserted for a lower byte write.

Or for a word write, assert both UDS and LDS

b. The memory or peripheral device addressed shall:

1. Decode the address on the address bus lines Al through A20.

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c. The CMB shall automatically assert DTACK logic low after a set time delay following AS going low.

d. The **MC68000** upon receipt of DTACK logic low shall:

1. Bring = and LDS logic high.

2. Bring = logic high.

3. **Remove all data from DO through D7 and D8** through **D15 by tristating its data lines by byte.**

4. Set WRITE logic high.

e. *TJe* **cMB DTACK circuit shall bring DTACK logic** high when AS goes high.

3.5.5.3 MC68000 interrupt handling

3.5.5.3.1 Interrupt encoding. Encoder TJ37 shall encode each interrupt of Table VI to a binary interrupt priority level per Table VI and output it to microprocessor U40.

TABLE VX. Interrupt encoding.

Intermpt	Decimal Interrupt Priority Level	Binary Encoded Interrupt Priority Level		
		$\overline{\text{IPL2}}$ (J1-39)	$\overline{\text{IPL1}}$ (J1-15)	$\overline{\text{IPL0}}$ (J1-40)
BIT TIMER INT	2	L	H	L
INTERRUPT 3 (PI-65)	3	**L**	**H**	**H**
WATCH XXX INT	4	**H**	**L**	**L**
SCP INT	5	**H**	**L**	**H**

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3.5.5.3.2 Interrunt decoding, acknowledgin and auto-vectoring. The encoded priority interrupt level shall be decoded by the microprocessor and compared to the present priority level in the status register of the microprocessor. If the incoming interrupt level is less than or equal to the current level in the status register, interrupt processing shall be postponed. When the interrupt priority level becomes higher than the current **priority le;el~ the microprocessor, per Figure .2, shall:**

a. Wait for the current instruction to complete.

b. Set function codes FCO, FC1 and FC2 all logic high per Table V.

c. Place the interrupt level on address lines Al, A2 and A3. Set A4 through A23 all logic high.

d. Assert address strobe AS (J1-32).

e. **Assert data strobes UDS (J1-8) and LDS (J1-33).**

f. **Internally generate a vector number based on the** interrupt priority level number, fetch the contents of the interrupt vector and load it into the program counter (autovectoring process) .

With the function codes all logic high, IACK shall be logic high to prevent selecting (enabling) on and off-board **memory and** the **generation of a DTACK (J1-1O) signal.**

3.5.5.4 Bus arbitration. A master-type device shall be capable of taking control of CMB buses away from microprocessor u40 by setting microprocessor u40's address, data and control lines to a high-impedance state per the following:

a. **with power applied to the CMB per 3.5.2, Dulling BUS REQUEST (J1-12) logic low shall Cause BUS Gm (J1-35) to go logic low within 0.528 xnicroseconds maximum.**

b. with BUS GRANT logic low and AS (J1-32) logic high, pulling BUS GRANT ACK (J1-36) logic low shall cause the following input/output signal lines of microprocessor U40 to be set to a high-impedance state which shall enable those CMB lines to be driven by an external master-type device.

Al to A20 (P2-17, 18, 55, 20, 19, 57, 59, 24, 58, 23, 60, 25, 26, 28, 62, 27, 37, 2, 36. and 1)

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The start-up BIT routine shall set (logic high) flags at the addresses of Table III for each item tested successfully. All flag bits of Table III shall be latched logic high following the otherwise, the flag bit shall be cleared (logic low). start-up BIT test.

TABLE III. start-up BIT test flags.

60022H		
60022H	D1	**EPROM** U10
60022H	D2	EPROM U1
60022H	D3	EPROM U2
60022H	D8	RAM u16
60022H	D9	RAM **U20**
60022H	**Dl 0**	**RAM U17**
60022H	**Dll**	**RAM U19**
60011H	**DO**	**ALU U4"**

Start-up BIT results shall be latched and disPlaYed on LED's DS4 through DS1 per Table IV, showing the CMB tested 'good communications with other system boards was not possible.

TABLE IV Start-urn BIT LED Status.

		CIRCUIT Board Represented
DS4	**Flashing at 4 Hz**	
DS3	**Flashing at 4 Hz**	
DS2	**Flashing at 4 Hz**	**Servo Controller Board (SCB)**
DS1	Lit	Computer and **Memory Board** CMB

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3.5.4.3 Manual reset curcuit. . with power aPPlied to the CMB per 3.5.2, bringing TEST RESET (J1-9) logic low **for at least** 1.4 microseconds shall cause POWER ON (P1-69), **RESET (P1-29), and HALT (J1-36) to go logic low, the microprocessor's address and data bus lines shall go to a high-impedance state and causing the microprocessor's control lines to deactivate per 3.5.4.1. When TEST RESET is allowed to return high, start-up BIT** *per* **3.5.4.2 shall run.**

3.5.4.4 circuit With power applied to the CMB per 3.5.2, bringing HALT (J1-38) logic low shall cause the microprocessor to stop executing instructions at the end of the current bus cycle, set its address *and* **data bus lines to a high-impedance state and render its control lines inactive per 3.5.4.1. Bringing HALT logic high shall cause the microprocessor to continue execution with the next bus cycle.**

*3.5.5 MC68000 microproces*sor **U40.**

3.5.5.1 The main MC68000 control signals shall be address strobe AS (J1-32),. YIRITE (J1-34) lower data strobe LDS (J1-331. upper data strobe UDS (Jl-8, data transfer acknowledge DTACK (J1-10) and bus error BERR (J1-14). During every bus cycle, the *MC68000* **shall output the address bits onto address lines Al through** *A20,* **then assert AS logic low to indicate there is a valid address on the address bus. A DTACK sign & shall normally initiate termination of the bus cycle. If tie DTACK signal is not received within 16 microseconds following AS going logic low, BERR shall go logic low to notify the MC68000 that a problem exists with the cycle under execution, therefore, terminate the cycle and setting the address and data bus lines to a high-impedance state. With BERR low, AS shall be released and UO high, AS shall go low, presetting countdown counter U53, and BERR shall go high. With BERR high, the MC68000 shall again attempt to rerun the same bus cycle by loading the same address on the address bus and asserting AS low (AS high). If a bTA~ signal is not received within 16 microseconds a second time, BERR shall again go logic low and halt microprocessor operation. The address and data buses shall be set to a high-impedance state and the control lines shall become inactive. The MC68000 shall only restart If It receives a logic low on TEST RESET (J1-9) for more than 1.4 microseconds or if CMB power Is removed then reapplied to operate the power-on reset circuit.**

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3.5.5.2 MC68000 read and write cycle

3.5.5.2.1 Read cycle during a single read bus cycle, microprocessor u40 shall received data from memory or a peripheral device and read it by bytes per the following sequence of events illustrated in Figure 2.

a. The MC68000 shall:

1. Set WRITE (JI-32) logic high to perform a *read* operation (READ, P1-24, shall be logic low).

2. Set function code outputs FCO, FCl and FC2 per Table V for the cycle type being executed.

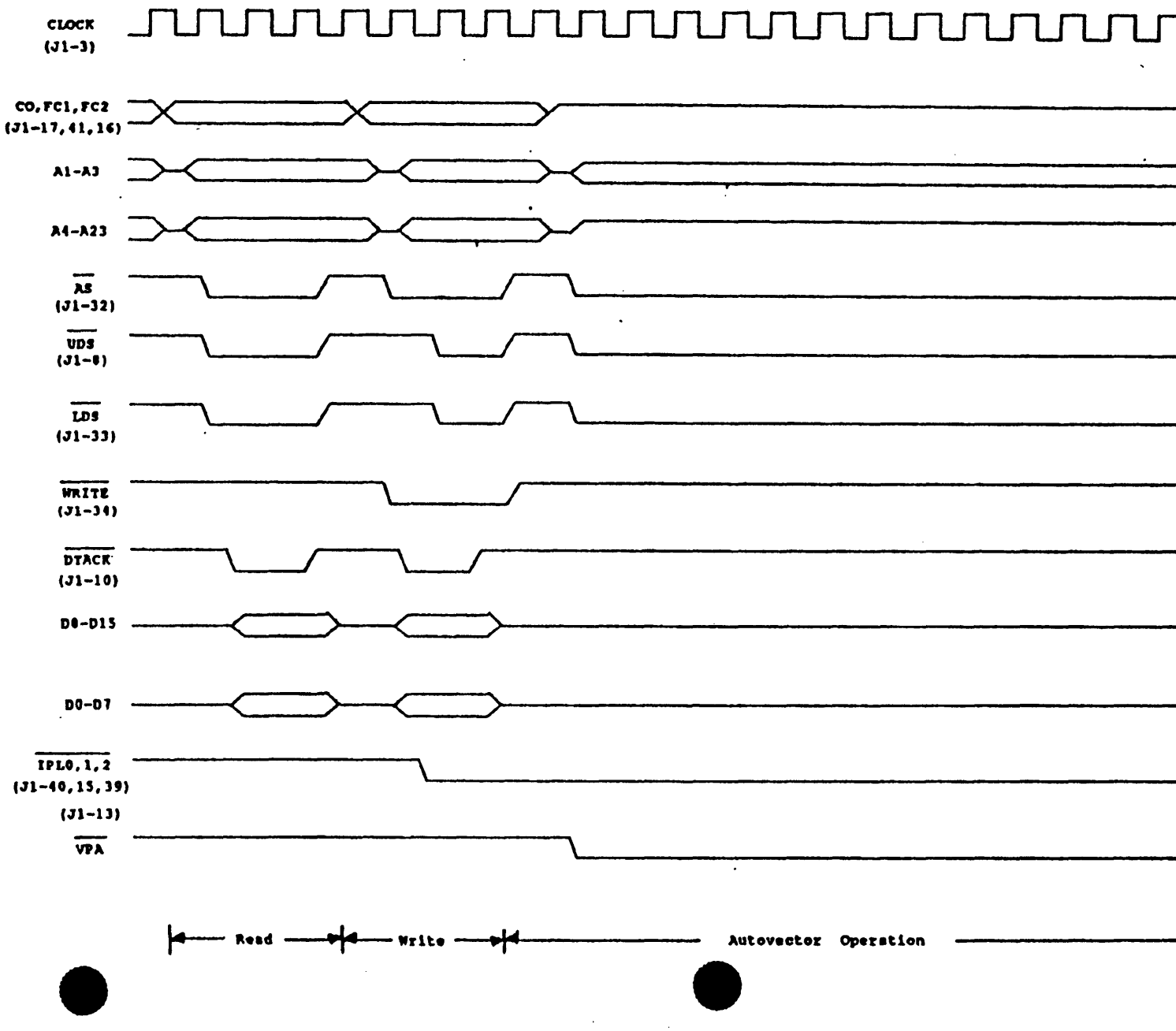
3. Place an address on address lines Al through A23.

4. Assert AS (J1-32) logic low.

5. Assert UDS (J1-8), LDS (J1-33) or both logic low, depending on whether the software instruction being executed requests an upper byte, lower byte or word read. If a byte read is requested, the internal AO bit is decoded as follows:

   For AO logic low, UDS is asserted.
   For AO logic high, LDS is asserted.

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TABLE V. **Function code output.**

Function Code Output			
FC2(J1-16)	FC1(J1-41)	FCO(J1-17)	cycle Type
L	L	L	(Undefined, Reserved)
L	L	H	User Data
L	H	L	User Program
L	H	H	(Undefined, Reserved)
H	L	L	(Undefined, Reserved)
H	L	H	Supervisor Data
H	H	L	Supervisor Program
H	H	H	Interrupt Acknowledge

b. The memory or peripheral device addressed shall:

1. Decode the address on the address bus lines Al *through A20.*

2. **Place data on data lines DO through D7 and D8** through D15 as required.

c. The CMB DTACK circuit shall automatically assert DTACK logic low after a set time delay following AS going low

d. The MC68000 shall upon receipt of DTACK low:

1. Latch the data on the data bus.

2. Negate UDS and LDSas asserted).

3. Negate AS.

e. The memory or peripheral device addressed shall remove the data from DO through D7 and D8 through D15 (as asserted) .

f. THEB DTACK circuit **shall bring** DTACK **logic high when** AS goes logic **high.**

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3.5.5.2.2 Write cycle. During a single write bus cycle, microprocessor U40 shall send data to a memory or peripheral device and write it by bytes per the following sequence of events; illustrated in Figure 2.

" a. The MC68000 shall:

1. Set function code outputs FCO, FCl and FC2 per Table v.

2. Place an address on address lines Al through A23.

3. Assert AS (J1-32) logic low.

4. Set WRITE (J>4) logic low to Perform a write operation (READ, Pi-24, shall be logic high)"

5. Place data on data lines DO through D7 and D8 through D15, depending on the AO bit as follows for a byte write:

For AO logic low, D8 through D15.
For AO logic high, DO through D7.
For a word write, data shall be placed on DO through D15 per the software instruction.

6. Assert UDS (J1-8) or LDS (J1-33) logic low depending on the AO bit as-follows:

For AO logic low, UDS shall be asserted for an upper byte write.

For AO logic high, LDS shall be asserted for a lower byte write.

or for a word write, assert both UDS and LDS

b. The memory or peripheral device addressed shall:

10 Decode the address on the address bus lines Al through A20.

2. Store data on DO through D7 if LDS is logic low, store data on D8 through D15 If UDS, is logic low or, store data on DO through D15 if LDS and UDS *are* both logic low*

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c. **The CMB shall automatically assert DTACK logic low after a set time delay following AS going low.**

d. **The** *MC68000* upon receipt **of DTACK** logic low shall:

1. Bring UDS and LDS logic high.

2. Bring AS logic high.

3. Remove all data from DO through D7 and D8 through D15 by tristating its data lines by byte.

4. Set **WRITE** logic high.

e. The CMB **DTACK** circuit shall bring DTACK logic high when AS goes **high.**

3.5.5.3 MC68000 interrupt handling.

3.5.5.3.1 Interrupt encoding. Encoder U37 shall encode each interrupt of Table VI to a binary interrupt priority level per Table VI and output it to microprocessor U40.

TABLE VI. Interrupt encoding.

Interrupt	Decimal Interrupt priority Level	Binary Encoded Interrupt Priority Level		
		IPL@ (J1-39)	IPL1 (J1"15)	IPLO (J1-40)
BIT TIMER INT	2	L	H	L
INTERRUPT 3 (PI-65)	3	L	H	H
WATCH DOG INT	4	H	L	L
SCP INT	5	H	L	H
INTERRUPT 6 (PI-26)	6	H	H	L

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3.5.5.3.2 Interrupt decoding, acknowleding and auto-vectoring. The encoded priority interrupt level shall be decoded by the microprocessor and compared to the present priority level in the status register *of* the microprocessor. **If** the incoming interrupt level is less than or equal to the current level in the status register, interrupt processing shall be postponed. When the interrupt priority level becomes higher than the current **priority** level, the microprocessor, per Figure .2, shall:

a. Wait for the current instruction to complete.

b. Set function codes FCO, FC1 and FC2 all logic high per Table V.

c. Place the interrupt level on address lines Al, A2 and A3. Set A4 through A23 all logic high.

d. Assert address strobe AS (J1-32).

e. Assert data strobes UDS (J1-8) and LDS (J1-33).

f. Internally generate a vector number based on the interrupt-priority level number, fetch the contents of the interrupt vector and load it into the program counter (autovectoring **Process) .**

With the function codes all logic high, IACK shall be logic high to prevent selecting (enabling) on and off-board **memory and** the **generation of a DTACK (J1-1O) signal.**

3.5.5.4 Bus arbitration. A master-type device shall be capable of taking control of CMB buses away from microprocessor u40 by setting microprocessor u40's address, data and control lines to a high-impedance state per the following:

a. **With power applied to the CMB per 3.5.2, Dulling ~ REQUEST (J1-12) logic low shall cause BUS GRANT- (J1-35) to go logic low within 0.528 microseconds maximum.**

b. **With BUS GRANT logic low and AS (J1-32) logic high, pulling BUS GRANT ACK {J1-36) logic low shall cause the** following input/output signal lines of microprocessor U40 to be set to a high-impedance state which shall enable those CMB lines to be driven by an external master-type device.

Al to **A20 (P2-17, 18, 55, 20, 19,** 57, 59, 24, 58, 23, **60, 25, 26, 28, 62, 27, 37,** 2, **36,** and 1)

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AS (J1-32)

$\overline{\text{WRITE}}$ (J1-34)

$\overline{\text{LDS}}$ (J1-33)

$\overline{\text{UDS}}$ (J1-8)

FCO (J1-17)

FCl (J1-41)

FC2 (J1-16)

DO to D15 (J1-44, 19, 20, 45, 5, 28, 4, 29, 21, 46, 22, 47, 30, 6, 31, and 7)

c. **With $\overline{\text{BUS GRANT ACK}}$ logic low, $\overline{\text{BUS GRANT}}$ should be negated.**

d. **$\overline{\text{BUS GRANT ACK}}$ should be negated when the external device has finished using the buses and control lines.**

3.5.6 peripheral control decoder U26 With power applied to the CMB per 3.5.2, peripheral control decoder U26 shall decode its input signals and output the signals shown in Table VII, during read and write bus cycles per 3.5.5.2.

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TABLE VII. Peripheral control decoder U26 operation.

Decoder Inputs						Output Decoded Logic Low	Resulting Output Logic High
A20 (P2-1)	IACK	$\overline{AS}$ (J1-32)	A19 (P2-36)	A18 (P2-2)	A17 (P2-37)		
X	H	X	X	X	X	None	None
X	X	H	X	X	X	None	None
H	X	X	X	X	X	None	None
L	L	L	L	L	L	$\overline{\text{PROM SELECT}}$ (Internal)	PROM SELECT (P2-63) SCP TEST(J1-18)
L	L	L	L	L	H	$\overline{\text{RAM SELECT}}$ (Internal)	RAM SELECT (J1-23)
L	L	L	L	H	L	$\overline{\text{SCP SELECT}}$ (Internal)	SCP TEST(J1-18)
L	L	L	L	H	H	$\overline{\text{STROBES}}$ (J1-48)	SCP TEST(J1-18)
L	L	L	H	L	L	None	None
L	L	L	H	L	H	$\overline{\text{I/O BOARD 1 SELECT}}$ (P1-27) $\overline{\text{I/O SELECT}}$ (J1-42)	None
L	L	L	H	H	L	$\overline{\text{I/O BOARD 2 SELECT}}$ (P1-66) $\overline{\text{I/O SELECT}}$ (J1-42)	None
L	L	L	H	H	H	None	None

Decoding $\overline{\text{SCP SELECT}}$ or $\overline{\text{I/O SELECT}}$ logic low shall cause $\overline{\text{OUT LD}}$ (J1-37) to go logic low when LDS is asserted, during a read or write bus cycle. Decoding SCP SELECT or I/O SELECT logic low shall cause $\overline{\text{OUT UD}}$ (J1-11) to go logic low when UDS is asserted, during a read or write bus cycle. Logic lows on $\overline{\text{OUT LD}}$ and $\overline{\text{OUT UD}}$ shall enable data bus transceivers U43 and U46, respectively, for reading from and writing to other FCP boards over bus D0 through D15 (P1-48, 8, 47, 7, 46, 6, 45, 5, 44, 4, 43, 3, 42, 2, 41 and 1).

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3.5.7 Address strobe delay circuit. The address strobe signal AS output by the microprocessor when address data is valid, shall be delayed by several time delays to provide Data Acknowledge (DTACK) signals to end each bus cycle. Different duration delays are required for reading and writing to RAM, reading IPROMSS and writing to and reading from peripheral devices. On a read cycle, DTACK is allowed to be asserted up to 90 nanoseconds before the data to be read is valid.

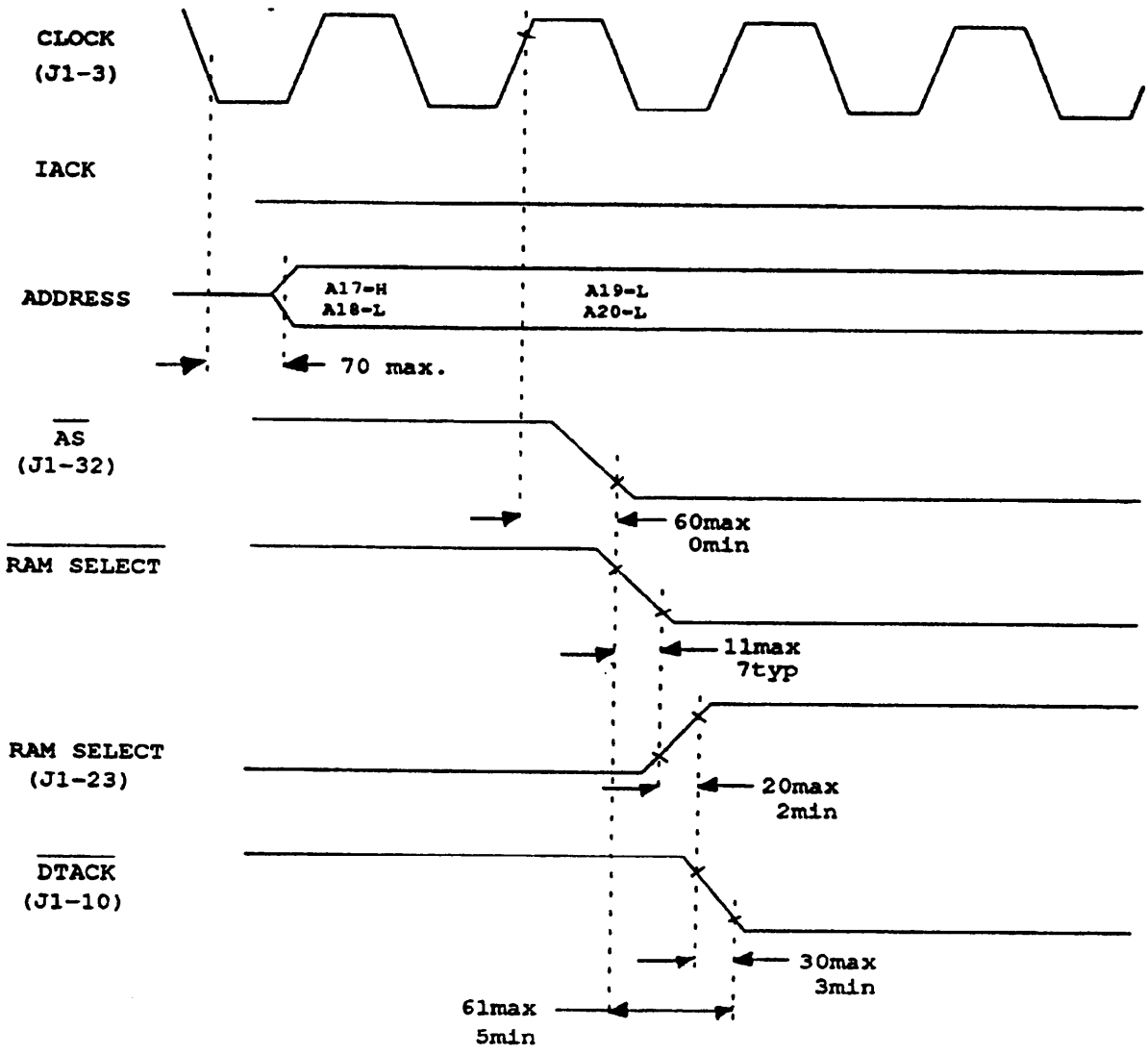
3.507.1 RAMS DTACK timing. With $\overline{\text{RAM SELECT}}$ decoded per 3.5.6, the microprecessor shall *receive a* DTACK (J1-1O) signal to end t~ bus cycle 5 to 61 nanoseconds after the microprocessor sets AS (J1-32) logic low. Timing shall be per *Figure* 3.

3.5*7.2 DTACK with PROM $\overline{\text{SELECT}}$ (P1-67) decoded per 3.5.6 during a read cycle. WRITE (J1-34) logic high, the microprocessor shall receive a DTACK (J1-1O) signal to end the bus cycle 132.5 to 228.5 nanoseconds after the microprocessor sets AS (J1-32) logic low. Timing shall be per Figure 4. DELAY 1 shall trigger DTACK.

3.5.7.3 Scp $\overline{\text{DTACK}}$ timing With $\overline{\text{SCP SELECT}}$ decoded per 3.5.6, during a write cycle WRITE (J1-34) logic low, the microprocessor shall receive a DTACK (J1-1O) signal to end the bus cycle 257.5 to 353.3 nanoseconds after the microprocessor sets AS (J1-32) logic low. timing shall be Per Figure 5. DELAY 2 shall trigger DTACK STROBES (J1-48) decoded per 3.5.6, instead of SCP SELECT, DTACK timing shall also be 257.5 to 353.5 nanoseconds.

3.5.7.4 EXTERNAL DTACK INPUT With $\overline{\text{1/0 BOARD 1}}$ or I/O BOARD 2 decoded per 3.5.6, or with 1 MEG SELECT decoded per Table VIII an external logic low DTACK signal input to the CMB at P1-12 shall cause the microprocessor or DTACK (J1-1O) input to go logic low within 5 to 48 nanoseconds after the logic low at P1-12 is applied.

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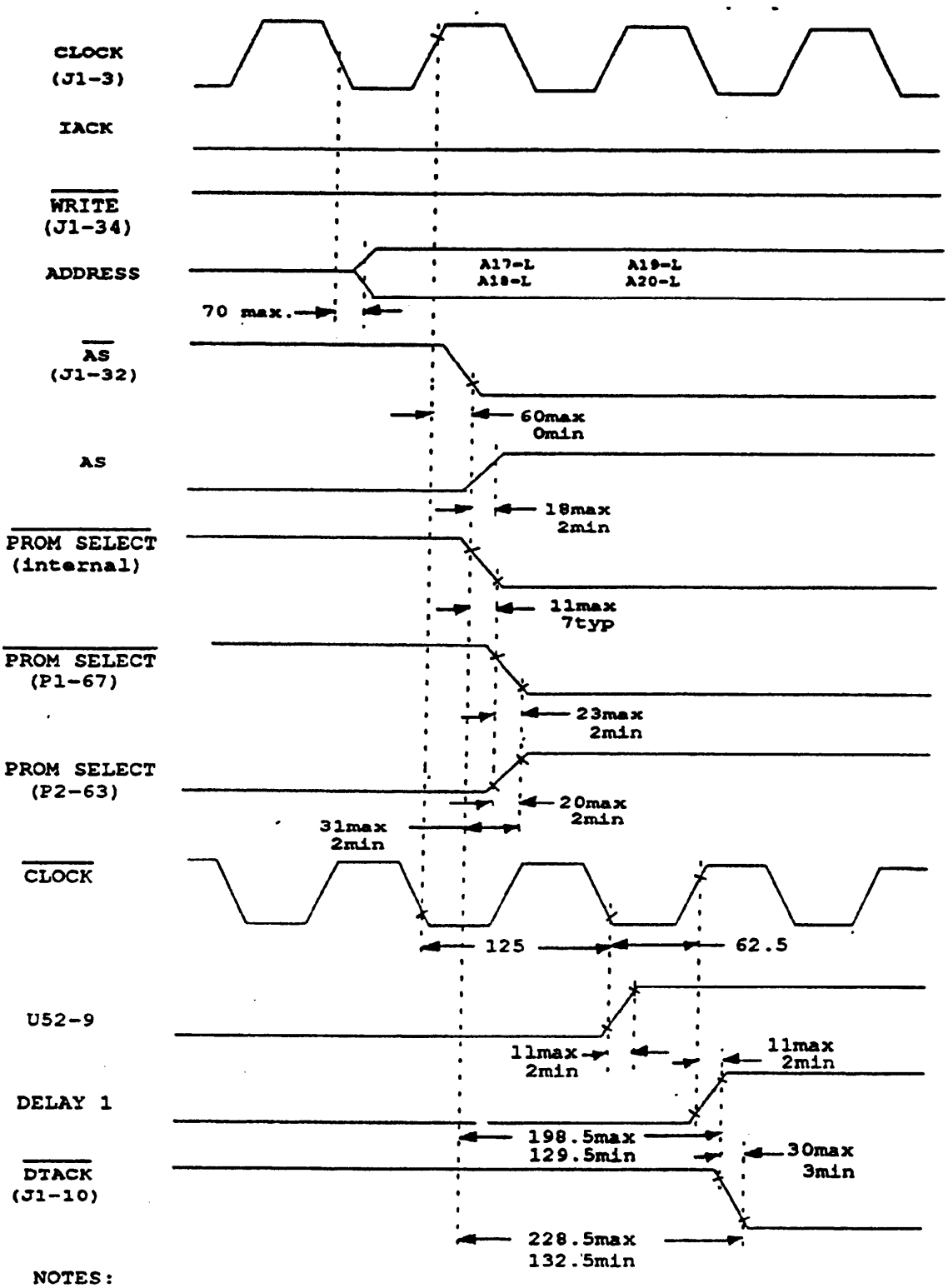


NOTES:

1. All times are in nanoseconds.
2. The timing reference levels are 0.8V for a logic low and 2.0V for a logic high.

FIGURE 3. RAM DTACK timing diagram.

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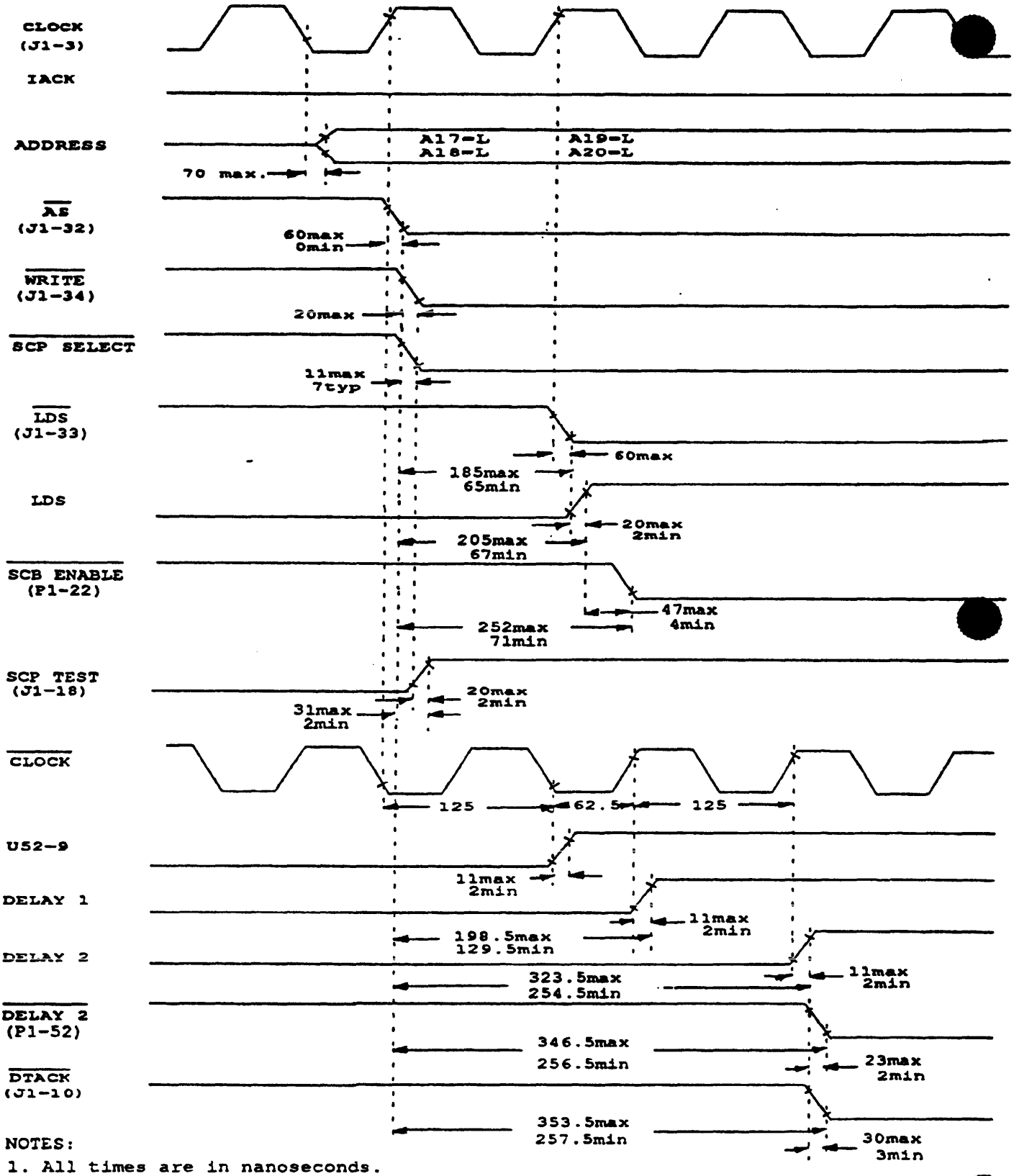


NOTES:

1. All times are in nanoseconds.
2. The timing reference levels are 0.8V for a logic low and 2.0V for a logic high.

FIGURE 4. PROM $\overline{\text{DTACK}}$ timing diagram.

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NOTES:

1. All times are in nanoseconds.
2. The timing reference levels are 0.8V for a logic low and 2.0V for a logic high.

FIGURE 5. SCP DTACK timing diagram.

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TABLE VIII. 1 MEG SELECT decodixg.

Inputs			Output
IACK	A2 0 (P2-1)	AS (J1-32)	1 MEG SELECT (=-31)
H	X	X	H
x	L	X	H
x	X	H	H
L	H	L	L

3 .5.7.5 DELAY READ timing. During a read cycle, = (Jl-34) logic high, DELAY READ (P2-12) shall go logi~high 131.5 to 223.5 nanoseconds after the microprocessor sets AS (J1-32) logic **low. DELAY 1** shall trigger DELAY-READ. This signal is used to delay reading the CMB timer (TIMER READ, P2-13), **or FCP status** (READ CARD SLOT, P2-4), whichever is decided by U24. Timing shall be per Figure 6.

3.5.7.6 2 DELAY WRITE pulse. 2 DELAY WRITE (Pi-64) shall go logic low for 297.5 to 483.5 nanoseconds, during a write cycle WRITE (J1-34) lo91c low, to provide the write pulse for writing to CONTROL (P2-8), LED DATA LOAD (P2-3), SCP INT CLEAR (P2-38) or TIMER WRITE (P2-45), whichever is decoded by U24. Timing shall be per Figure 7.

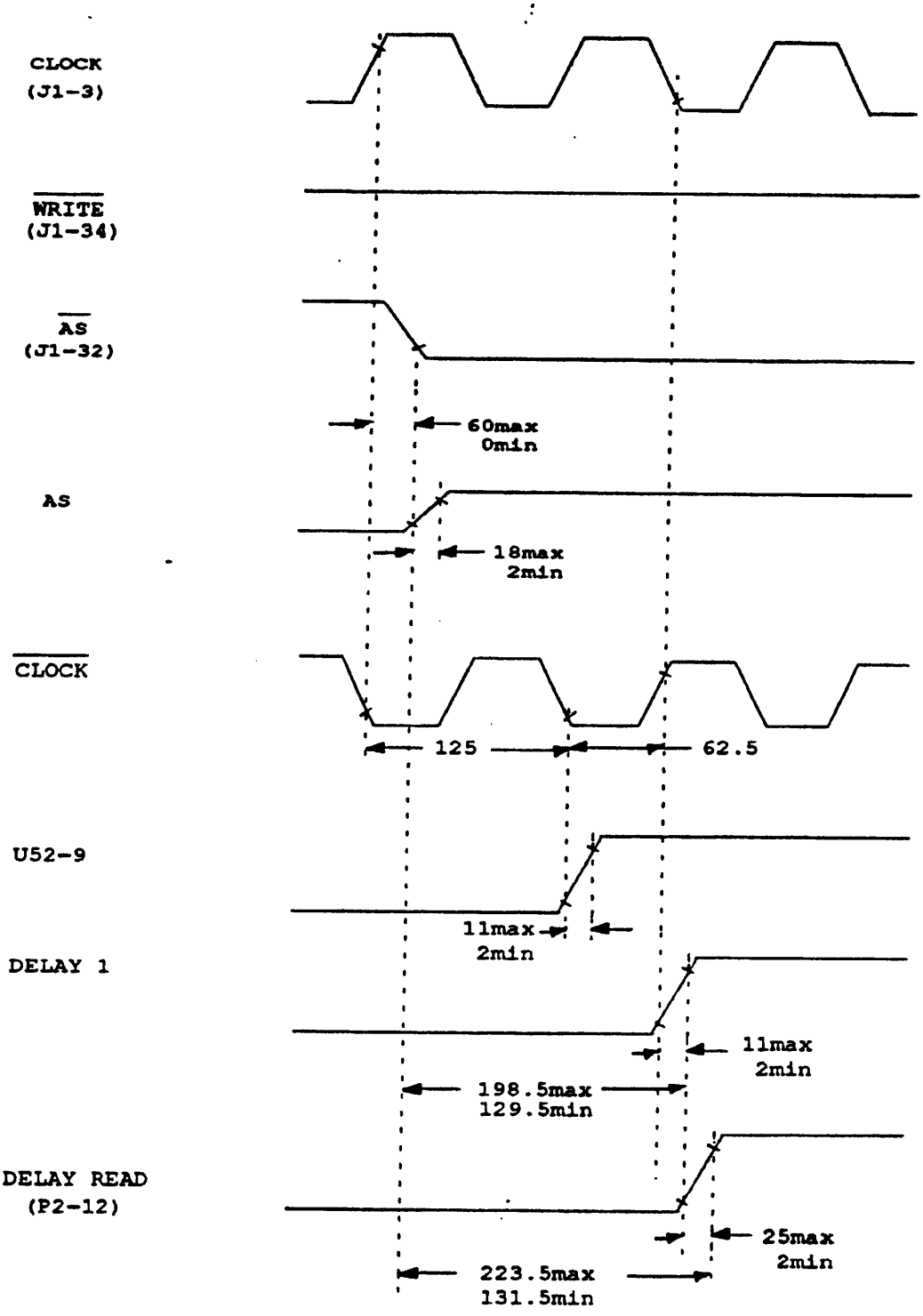
3.5.8 Write/read control decoder U24. Write and read control functions shall be decoded by 1/0 control decoder U24, which shall be enabled by STROBES (J1-48) from peripheral control decoder U26 per 3.5.6. The write and read control signals shall decode logic low when supplied with the inputs of Tables IX and x, respectively.

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TABKE IX X/O control decoder U24 "write" control signals.

U24 Inputs				
A16 (P2-27)	A15 (P2-62)	STROBES (J1-48)	2 DELAY WRITE	u24 Decodecl Output
L	L	L	L	CONTROL (P2-8)
L	H	L	L	LED DATA LOAD (P2-3)
H	L	L	L	SCP INT CLEAR (P2-38)
H	H	L	L	TIMER WRITE (P2-45)
x	x	H	x	None
x	x	x	H	None

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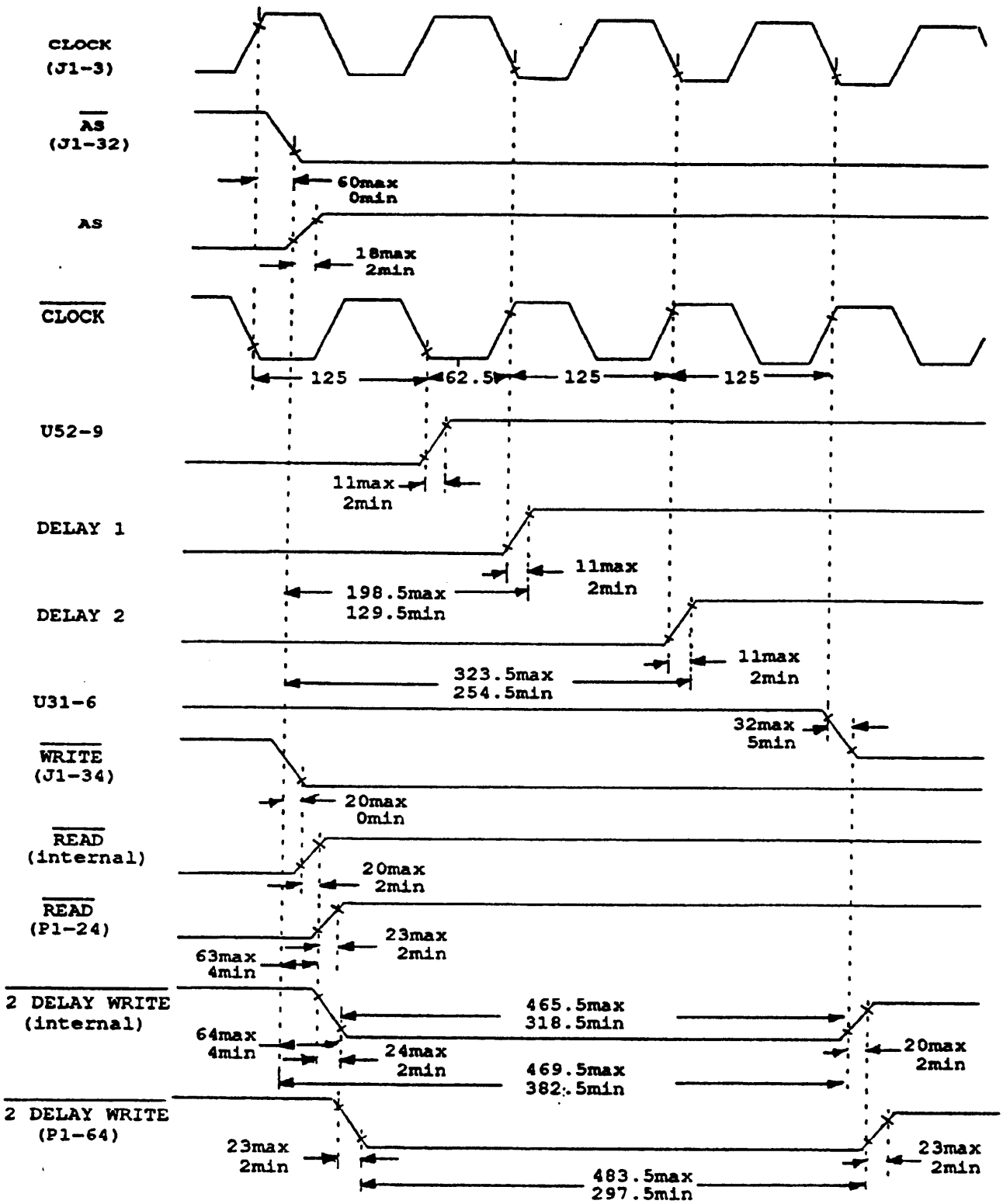


NOTES:

1. All times are in nanoseconds.
2. The timing reference levels are 0.8V for a logic low and 2.0V for a logic high.

FIGURE 6. DELAY READ timing diagram.

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NOTES:

1. All times are in nanoseconds.
2. The timing reference levels are 0.8V for a logic low and 2.0V for a logic high.

FIGURE 7. $\overline{2}$ DELAY WRITE timing diagram.

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TABLE X. I/O control decoder U24 "read" control Signals.

U24 Xxl!puta				
Al 6 (P2-27)	Al 5 (P2-62)	STROBES (J1-48)	DSLAY READ (P2-12)	u24 Decoded Output
L	L	L	H	N/C
L	H	L	H	N/C
H	L	L	H	READ CARD SLOT (P2-4)
H	H	L	H	TIMER READ (P2-13)
X	X	H	X	None
X	X	X	L	None

3.5.9 Programmable timer circuit operation.

3.5.9.1 Timer control latch U18. Decoding CONTROL (P2-8) per 3.5.8 shall enable timer control latch u18. The outputs of u18 shall be selected by address lines A12 through A10 per Table XI, while U18 is enabled. The data on data line DO (J1-44) shall write to the decoded output.

TABLE XZ. Timer control latch u18 decode.

U18 Inputs				
CONTROL (P2-8)	A12 (P2-25)	All (P2-60)	Al 0 (P2-23)	U18 - Decoded Output
L	L	L	L	NIC
L	L	L	H	N/C
L	L	H	L	N/C
L	L	H	H	BIT TIMER GATE ON (P2-9)
L	H	L	L	BIT TIMER INT ON (P2-41)
L	H	L	H	WATCH DOG GATE ON (P2-44)
L	H	H	L	WATCH DOG INT ON (P2-7)
L	H	H	H	SCP INT ON (P2-42)
H	X	X	X	*None*

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3.5.9.2 Timer address. Timer address latch U12 shall latch address A14 (P2-28) onto Al TIMER (P2-14) and latch address A13 (P2-26) onto AO TIMER (P2-15) when TIMER READ (P2-13) is decoded per 3.5.8. With TIMER read logic high, Al TIMER shall track changes in A14 and AO TIMER shall track changes in A13.

3*5.9*3 Timer U15.. Timer U15 shall contain three counters which shall output interrupts to the MC68000 microprocessor after programmed time delays per the following paragraphs.

3.5.9.3.1 BIT timer inerrupt.. The output of counter 0, OUT 0 (P2-11), is connected to the Input of counter 1 to form the BIT timer. Counter 0 is clocked by LS CLOCK (8 MHz) and shall count continuously because its gate is tied logic high. Each counter shall be programmed with a control byte and initial count . Counter 0 shall be programmed to mode 3 per Figure 8, the square wave mode, where an initial count of N shall result In a square wave of period N.. Counter 1 shall be programmed to mode 0 per Figure 8, the interrupt *on* terminal count mode, and its output, OUT 1 (P2-43), shall go logic high when the counter has counted down in binary fashion to zero from the programmed initial count. Counter 0 shall be loaded with the number of 8 MHz cycles to count and counter 1 shall be loaded with the number of counts to decrement. The BIT timer interrupt shall be programmable for delays of 125 nanoseconds to 8.9 minutes. BIT TIMER INT shall go logic low after the programmed time delay and generate a priority level 2 Interrupt per 3.5.5.3.1, when the following commands are written per 3.5.5.2.2:

a. Write control byte 36H (hexadecimal) for counter 0 onto upper data byte (D15-~e= 07EoooH~ This address shall decode TIMER WRITE (P2-45) logic low per 3.5.8.

b. For counter 0, write least significant byte of ln~t~al count onto upper data byte at address 078000H, then most significant byte of initial count onto the upper data byte at address 078000H.

c. Write control byte 70H for counter 1 onto the upper data byte at address 07EOOOH.

d. P'or counter 1, write least significant byte of initial count onto upper data byte at address 07AOOOH, then most significant byte of initial count onto the upper data byte at address 07EOOOH.

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D15	D14	D13	D12	D11	D10	D9	D8
SC1	SC0	RW1	RW0	M2	M1	M0	BCD

SC-Select Counter

SC1	SC0	
L	L	Select Counter 0
L	H	Select Counter 1
H	L	Select Counter 2
H	H	Read-Back Command

RW-Read/Write

RW1	RW0	
L	L	Counter Latch Command
L	H	Read/Write Least Significant Byte Only
H	L	Read/Write Most Significant Byte Only
H	H	Read/Write Least Significant Byte First, Then Most Significant Byte

M-Mode

M2	M1	M0	
L	L	L	Mode 0
L	L	H	Mode 1
X	H	L	Mode 2
X	H	H	Mode 3
H	L	L	Mode 4
H	L	H	Mode 5

BCD

L	Binary Counter (16 bits)
H	Binary Coded Decimal (BCD) Counter (4 Decades)

FIGURE 8. Counter control byte format.

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e. **Write DO (J1-44) logic high onto lower data byte at address 061401H to latch BIT TIMER INT ON (P2-41) logic high, i.e., to ready interrupt gate U21 for the BIT timer interrupt.**

f. Write DO (J1-44) logic high onto lower data byte at address 060C01H to latch BIT TIMER GATE ON (P2-9) logic high to count.

3.5.9.3.2 WATCH DOG timer interrupt function. Counter 2 of U15 is the WATCH DOG timer. Counter 2 is clocked by a 500 KHz clock and shall be programmed with a control byte and initial count. Counter 2 shall be programmed to mode 0 per Figure 8, the interrupt on terminal count mode, and its output, OUT 2 (P2-10), shall go logic high when the counter has counted down in binary fashion to zero from the programmed initial count. Counter 2 shall be loaded with the number of counts to decrement. The WATCH DOG timer interrupt shall be programmable for delays of 2.0 microseconds to 131 milliseconds. WATCH DOG INT shall go logic low after the programmed time delay and generate a priority **level 4 interrupt per 3.5.5.3.1, when the following commands are written per 3.5.5.2.2:**

a. Write control byte BOH (hexadecimal) for counter 2 onto upper data byte (D15-D8) at address 07E000H. This address shall decode TIMER WRITE (P2-45) logic low per 3.5.8.

b. **For counter 2, write least significant byte of initial count onto upper data byte at address 07C000H, then most significant byte of initial count onto upper data byte at address 07C000H.**

c. **Write DO (J1-44) logic high onto lower data ~te at address 061801H to latch WATCH DOG INT ON (P2-7) logic high, i.e., to ready interrupt gate U21 for the WATCH DOG timer interrupt.**

d. **Write DO (J1-44)" logic high onto data byte at address 061401H to latch WATCH DOG GATE ON (P2-44) logic high to count .**

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3.5.9.3.3 Timer read function. **Timer U15 shall latch the status and count of each of its three counters when a Read-Back Command per Figure 9 is sent to the timer. Decoding TIMER READ (P2-13) logic low per 3.5.8 for each counter, shall enable the status per** Figure 10 and count of each counter to be read; per programmed Read-Back Command. If count was requested, the first read shall return the least significant byte of the count and the second read shall return the most significant byte of the count for the addressed counter. If status and count were both requested, the first read shall return the status byte, the second read shall return the least significant byte of the count and the third read shall return the most significant byte of the count for the address counter. Write and read addresses shall be as follows:

a. The **Read-Back Command of** Figure **9** shall be written to address 07E000H (hexadecimal) Per 3.5.5.2.2.

b. **Counter O shall be read at address 078000H per 3.5.5.2.1.**
Counter 1 shall be read at address 07A000H per 3.5.5.2.1.
Counter 2 shall be read at address 07C000H per 3.5.5.2.1.

3.5.10 SCP interrupt. **A** 64 **Hz square wave** TTL interrupt signal shall be received **at CPU INTERRUPT (Pi-63) from the FCP servo controller processor for establishing communications with the cMB processor. With power applied to the CMB per 3.5.2, SCp** FF (P2-6) shall go and remain logic high on the trailing edge of the first received CPU INTERRUPT pulse. With SCP INT ON (P2-42) decoded per 3.5.9.1, SCP INT shall go logic low and generate interrupt priority level 5 per 3.5.5.3.1. When SCP lNT CLEAR (P2-38) is decoded per 3.5.8, SCP FF shall go logic low and SCP INT shall go logic high and remove interrupt priority level 5.

3.5.11 FCP status buffer U32. Decoding **READ CARD SLOT (P2-4) per 3.5.8, shall enable inverting FCP status buffer U32 to transfer the input signals of Table XII onto data lines D7** through D4 per Table XII.

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D15	D14	D13	D12	D11	D10	D9	D8
H	H	$\overline{\text{COUNT}}$	$\overline{\text{STATUS}}$	CNT2	CNT1	CNT0	L

NOTES:

1. D15 and D14 shall both be high to specify the Read-Back Command.
2. D13 shall be low to request the count of selected counter.
3. D12 shall be low to request the status byte of selected counter, Figure 10.
4. D11, D10 and D9 shall be high to select the counter(s) to be read.
5. D8 shall be low to select binary count.

FIGURE 9. Read-Back Command format.

D15	D14	D13	D12	D11	D10	D9	D8
OUTPUT	NULL COUNT	RW1	RW0	M2	M1	M0	BCD

NOTES:

1. D15 shall be high if OUT pin of counter selected is high.
D15 shall be low if OUT pin of counter selected is low.
2. D14 shall be high if last count written to counter register has not been loaded into the counting element.
D14 shall be low if last count written to counter register has been loaded into the counting element.
3. D13 through D8 shall be the same as in Figure 8.

FIGURE 10. Status byte format.

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TABLE XII. FCP Buffer status.

Buffer U32 Input Signal	Data Line	Signal DATA Line
31.250 ±2 ± 5 Hz. TTL square wave	X)7 (J1-29)	Inverted 31.25 KHz ± 5 Hz. TTL sguare wave
CPU INTERRUPT (Pi-63) 64 +/- 1 Hz TTL square wave	D6 (J1-4)	Inverted 64 Hz i 1 Hz. TTL Square wave
1/0 BOARD 1 IN (P1-11) logic low [Grounded when IOB of FCP is installed into FCP card cage]	D5 (J1-28)	Logic high
SCB BOARD IN (Pi-23) logic low [Grounded when SCB of FCP is installed into FCP card cage]	D4 (J1-S)	Logic high

3.5.12 BIT latch U25 and CO with power amlied to the CMB per 3.5.2 and with LED DATA LOAD ;P2-3) repeatedly decoded per 3.5.8 at a rate of 10 Hertz or greater, CPU RUN (J1-43) shall go and remain logic high. with CPU RUN logic high:

a. BIT latch U25 shall latch the lines D3-DO every time it is clocked by LED DATA LOAD. The latched outputs shall illuminate or extinguish LEDs DS1-DS4 as follows:

With D3 (J1-45) logic high . . . .DSl Shall be On
With D3 (J1-45) logic 10W . . . .DS1 shall be off
with D2 (J1-20) logic high. .DS2DS2 shall be On
with D2 (J1-20) logic low.- ..* . . . .DS2 shall be off
with D1 (J1-19) logic high . . . .DS3 shall be On
with D1 (J1-19) logic 10W . . . .DS3 shall be off
With DO (J1-44) logic high . . . .DS4 shall On
with DO (J1-44) logic 10W . . . .DS4 shall be off

b. COMPUTER CYCLE (P2-35, 70) shall go logic low to +24 V (BTRY) RTN P2-34: I.e., external lamp illuminates Discontinuing the LED DATA LOAD pulses shall cause CPU RUN to go logic low after 96 ± 10 miliseconds. then EDS DS1-DS4 shall extinguish and COMPUTER CYCLE shall go logic high; i.e., external lamp extinguishes.

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3.5.13 Random Access MemorY (RAM) circuit. CMB RAM memory presently consists of four 4K x 4 bit chips; U19, U17, u20 and **U16. There are Provisions** *for* **adding four more 85 nanosecond access** time **4K** X-4 **bit** chips; U62, U61, U63 **and u60. U19 and u17** form the low byte and chips u20 and u16 **form the high byte.**

3.5.13.1 RAM chin selection. RAM **chip selection shall be** by byte per Table XIII.

TABLE XXII. RAM byte selection.

1 3 (P2-26)	RAM SELECT (per 3.5.6)	LDS J1-33)	UDS (S1-8)	RAM Byte-Enab1ing Signal Decoded
L	L	L	x	**RAM** (J1-25) Existing RAM Lower Data BYte
L	L	x	L	RAM (J1-50) Existing RAM Upper Data Byte
H	L	L	x	RAM (JI-24) Future RAM Lower Data Byte
H	L	x	**L**	RAM (J1-49) Future RAM Upper Data Byte

3.5.13.2 RAM write operation. During a write bus cycle per 3.5.5.2.2, WRITE (J1-34) shall be logic low and enable the RAM wte **selected per 3.5.13.1. Data on-data lines D15 through D8 and D7 through DO shall write to RAM depending upon whether the upper, lower or both data strobes were asserted.**

3.5.13.3 RAM read operation. During a read bus cycle per 3.5.5.2.1, WRITE (J1-34) shall be logic high to enable the RAM byte selected per 3.5.13.1 to output data onto the data bus. The data read from RAM shall be output onto D15 through D8 **and D7** through DO depending on whether the upper, lower or **both data** s t robes were asserted.

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3.5.14 BOOT ON circuit. With power applied to the CMB per 3.5.2, PROM SELECT (PI-67) decoded per 3.5.6 and a logic low applied to BOOT ON (PI-25) shall:

a. cause BOOT TWO (P2-40) to go logic low and, during a read or write bus cycle per 3.5.5.2.

(1) OUT LD (J1-37) shall go logic low when LDS (J1-33) is logic low.

(2) OUT UD (J1-11) shall go logic low when UDS (J1-8) is logic low.

OUT LD and OUT UD- shall enable lower byte data transceiver, U43 and upper byte data transceiver U46, respectively, to connect the CMB internal data bus to CMB edge connector PI to allow the MC68000 on the CMB to execute programs in ROM external to the CMB. Address buffers U38 and U41 shall bring address llnes Al through A16 to edge connector PI.

b. **Prevent on board EPROMS from loading data onto the data bus by making PROM UD (P2-5) and PROM LD (P2-39) logic** high.

3.5.15 Erasable Prosrammable Read Only Memo ry (EPROM) circuit. CMB EPROM memory shall consist of four 32K x 8 bit chips; **Ul, U2, U10 and U11. Each EPROM check sum shall be per drawing 12563341.**

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3.5.15.1 EPROM chip selection. EPROM chip selection (enabling) shall be per Table XIV.

TABLE XIV. EPROM chip selection and data transceiver enabling signals.

Inputs					EPROM	EPROM DATA
$\overline{\text{BOOT}}$ $\overline{\text{ON}}$ (P1-25)	A16 (P2-27)	$\overline{\text{PROM}}$ $\overline{\text{SELECT}}$ (per 3.5.6)	$\overline{\text{LDS}}$ (J1-33)	$\overline{\text{UDS}}$ (J1-8)	SELECTION U3 decoder output (Logic Low)	Transceiver Enabling Signal Signal (Logic Low)
H	L	L	L	X	$\overline{\text{PROM OL}}$ (P2-68), U10	$\overline{\text{PROM LD}}$ (P2-39)
H	H	L	X	L	$\overline{\text{PROM OU}}$ (P2-65), U11	$\overline{\text{PROM UD}}$ (P2-5)
H	H	L	L	X	$\overline{\text{PROM IL}}$ (P2-67), U1	$\overline{\text{PROM LD}}$ (P2-39)
H	H	L	X	L	$\overline{\text{PROM IU}}$ (P2-30), U2	$\overline{\text{PROM UD}}$ (P2-5)

3.5.15.2 EPROM read operation. During a read cycle per 3.5.5.2.1, $\overline{\text{READ}}$ (P1-24) shall go logic low and cause EPROM READ (P2-22) to go logic low. Address lines A1 through A15, through address buffers U9 and U8, shall address memory locations in the four on-board EPROMs. When AS, then LDS and UDS are asserted, the EPROM chips selected per 3.5.15.1 shall output the data in their addressed locations through data receivers U13 and U14 for the low and high data bytes, respectively, to CMB data bus lines D0 through D7 and D8 through D15. The data transceivers shall be enabled by PROM LD and PROM UD per 3.5.15.1.

3.5.16 Input/output requirements. All inputs, outputs and test points on the CMB shall be standard transistor-transistor logic (TTL) levels except the computer cycling lamp output. This output on P2 pins 35 and 70 shall be connected to a 24 volt 40 mA lamp which has its other side connected to +24 V. The minimum and maximum input and output TTL voltage levels shall be as follows:

	MIN	TYP	MAX	
VOL	-	0.2	0.4	VOLTAGE OUTPUT LOW
VOH	2.4	3.4	-	VOLTAGE OUTPUT HIGH
VIL	-	-	0.7	VOLTAGE INPUT LOW
VIH	2.0	-	-	VOLTAGE INPUT HIGH

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3.5.16.1 Computer memory board inputs. The C14B inputs shall be as listed in Table XV. Unless-otherwise noted in the table, all signal levels are as specified in 3.5.16.

3.5.16.2 Computer memory board outputs. The CMB outputs shall be as listed in Table XVI. The outputs shall have no direct relationship to the inputs but shall be dependent upon the CMB software which dictates the type of cycle the MC68000 CPU is executing. The exceptions to this shall be RESET, Power ON, 500 KHZ, **and LS CLOCK (8 MHZ). RESET and POWER ON shall only be generated when +5 Vdc is initially applied to the CMB. The 500 KHz and LS CLOCK (8 MHZ) shall be generated whenever the 8 MHZ clock is not gated off. Unless otherwise noted in the table, all signal levels are TTL as specified in 3.5.16.**

3.5.16.3 Computer memory board inPuts/outputs. The CMB inputs/outputs shall be as listed in Table XVII. Unless otherwise noted in the table, all signal levels are TTL **as** specified in 3.5.16.

3.5.16.4 Computer memory board test points. The CMB shall contain test points as listed in Table XVIII. Some of the test points shall be both inputs and outputs. If the CPU is permanently reset, which sets its busses and control lines to a high-impedance state, then these test points shall be inputs. If the CPU is running, then these test points shall be outputs. Unless otherwise noted in the table, all signal levels are TTL as specified in 3.5.16.

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TABLE XV. Computer memory board inputs.

CONNECTOR PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-9	+5 V RTN	GROUND
P1-11	$\overline{\text{I/O BOARD 1 IN}}$	
P1-12	$\overline{\text{DTACK}}$	
P1-18	+5 V RTN	GROUND
P1-23	$\overline{\text{SCB BOARD IN}}$	
P1-25	$\overline{\text{BOOT ON}}$	
P1-26	$\overline{\text{INTERRUPT 6}}$	
P1-32	+5 V	+5 V ± .25 Vdc
P1-33	+5 V	+5 V ± .25 Vdc
P1-34	+5 V	+5 V ± .25 Vdc
P1-35	+5 V RTN	GROUND
P1-36	+5 V RTN	GROUND
P1-37	+5 V RTN	GROUND
P1-38	±15 V RTN	NOT USED ON CMB
P1-39	-15 V	NOT USED ON CMB
P1-40	+15 V	NOT USED ON CMB
P1-49	+5 V RTN	GROUND
P1-58	+5 V RTN	GROUND
P1-63	CPU INTERRUPT	
P1-65	$\overline{\text{INTERRUPT 3}}$	
P1-72	+5 V	+5 V ± .25 Vdc
P1-73	+5 V	+5 V ± .25 Vdc
P1-74	+5 V	+5 V ± .25 Vdc
P1-75	+5 V RTN	GROUND
P1-76	+5 V RTN	GROUND

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TABLE XV. C-uter memory board inputs (continued)

connector Pin	signal name	DeSCRIPTIVB dATA
PI-77	+5 V RTN	GROUND
PI-78	*15 V RTN	NOT USED ON CMB
Pi-79	-15 v	NOT USED ON CMB
PI-80	+15 v	NOT USED ON CMB
P2-32	+5 v	+5 V ± .25 vdC to EPROM
P2-33	+5 v	+5 V ± .25 Vdc TO EPROM

TABLE XVI. computer memory board outputs.

CONNECTOR PIN	SXGNAL NAME	DESCRIPTIVE DATA
PI-13	A16	
PI-14	A14	
PI-15	A12	
PI-16	A10	
PI-17	A8	
P1-19	A6	
P1-20	A4	
P1-21	AZ	
PI-22	$\overline{\text{SCB}}$ ENABLE	
Pi-24	E	
PI-27	$\overline{\text{1/0 BOARD 1 SELECT}}$	
PI-28	E	
PI-29	RESET	
P1-50	CLOCK (500 KHz)	
PI-52	$\overline{\text{DELAY 2}}$	
Pi-53	Al 5	
I-54	A13	

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TABLB **M. Computer memory board** outputs. (Continuad)

CONNBCTOR Pin	sIGnAL MANE	DESCRIPTIVE DATA
PI-55	Al 1	
PI-56	**A9**	
PI-57	**A7**	
Pi-59	**AS**	
P1-60	**A3**	
P1-61	**Al**	
PI-64	**2 DELAY WRITE**	
PI-66	**1/0 BOARD 2 SELECT**	
Pi-67	**PROM SELECT**	
Pi-68	**E**	
PI-69	**POWER ON**	
P1-70	**LS CLOCK (8 MHz)**	
P2-29	A18	
P2-31	1 MEG SELECT	
P2-34	+24 V (BTRY) RTN	+24 Vdc RTN
P2-35	COMPUTER CYCLE	DRIVES 24 V 40 MA LAMP
P2-64	A17	
P2-66	A19	
P2-70	COMPUTER CYCLE	DRIVES 24 V 40 MA LAMP

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TABLE XVII. Computor memor'y board inputsOutputs.

connector Pin	SiGNAl NAMe	DESCRIPTIVE data
P1-1	D15	
P1-2	D13	
P1-3	D11	
P1-4	D9	
PI-5	D7	
PI-6	D5	
PI-7	d3	
PI-8	Dl	
PI-41	D14	
PI-42	D12	
PI-43	D10	
Pi-44	D8	
Pi-45	D6	
Pi-46	D4	
Pi-47	D2	
Pi-48	B1/2 DO	

TABLE XVIII. Computer memo= board test pOints.

CONNECTOR PIN	SIGNAL NAME	DESCRIPTIVE DATA
P2-1	A20	
P2-2	A18	
P2-3	LED DATA LOAD	
P2-4	READ CARD SLOT	
P2-5	PROM UD	
P2-6	SCP F	
P2-7	WATCH DOG INT ON	
P2-8	CONTROL	
P2-9	BIT TIMER GATE ON	

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TABLE XVIII. Computer memory board taut Points. (Continued)

connector Pin	Signal name	DESCRIPTIVE DATA
P2-10	OUT 2	
P2-11	OUT o	
P2-12	DELAY READ	
P2-13	TIMER READ	
P2-14	A1 TIMER	
P2-15	AO TIMER	
P2-17		
P2-18	A2	
P2-19	AS	
P2-20	A4	
P2-22	EPROM READ	
P2-23		
P2-24	A8	
P2-25	A12	
P2-26	A13	
P2-27		
P2-28	A14	
P2-30	PROM IU	
P2-36	Al 9	
P2-37	A17	
P2-38	SCP INT CLEAR	
P2-39	PROM LD	
P2-40	BOOT TWO	
P2-41	BIT TIMER INT ON	
P2-42	SCP INT ON	
P2-43	IOUT1	

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TABLE XVIII. Computer memory board test points, (Continued)

connector PIN	sigNAL NAmE	DeSCRIPTiVB DATA
P2-44	WATCHDOG GATE ON	
P2-45	$\overline{\text{TIMER WRITE}}$	
P2-55	A3	
P2-57	A6	
P2-58	A9	
P2-59	A7	
P2-60	All	
P2 -62	A15	
P2-63	$\overline{\text{PROM SELECT}}$	
P2-65	$\overline{\text{PROM OU}}$	
P2-67	$\overline{\text{PROM IL}}$	
P2-68	P R O M	
J1-1	$\overline{\text{LS CLOCK OFF}}$	
J1-2	$\overline{\text{INTERNAL CLOCK DISABLE}}$	
J1-3	CLOCK	
J1-4	D6	
J1-5	D4	
J1-6	D13	
J1-7	D15	
J1-8	UDS	
J1-9	$\overline{\text{TEST RESET}}$	
J1-10	$\overline{\text{DTACK}}$	
J1-11	-	

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TABLE XVIII. Commput ter memory board test points (continuedl

CONNECTOR PIN	SIGNAL NAMB	DESCRIPTIVE DATA
J1-12	BUS REQUEST	
J1-13	E	
J1-14	m	
J1-15	m	
J1-16	FC2	
J1-17	FCO	
J1-18	SCP TEST	
J1-19	D1	
J1-20	D2	
J1-21	D8	
J1-22	D10	
J1-23	RAM SELECT	
J1-24	RAM IL	
J1-25	RAM OL	
J1-26	LS CLOCK IN	
J1-27	EXT CLOCK	
J1-28	D5	
J1-29	D7	
J1-30	D12	
J1-31	D14	
J1-32	AS	
J1-33	LDS	
J1-34	WRITE	

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TABLE XVIII. Computer memory board test points (Continued)

CONNECTOR Pin	SIGNAL NAME	DESCRIPTIVE DATA
J1-35	BUS GRANT	
J1-36	BUS GRANT ACK	
J1-37	OUT LD	
J1-38	HALT	
J1-39	IPL-2	
J1-40	IPL 0	
J1-41	FC 1	
J1-42	I/0 SELECT	
J1-43	CPU RUN	
J1-44	DO	
J1-45	D3	
J1-46	D9	
J1-47	D11	
J1-48	STROBES	
J1-49	R A M	
J1-50	RAM OU	

3.5.16.5 Computer memory board unused connector pins. The CMB unused connector pins shall be as listed in Table XIX.

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be *in* accordance with the general requirements of Note 2 of Drawing 12950877.

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TABLE xxx. Computer memo- board snares.

CONNECTOR PIN
PI-IO
P1-30
PI-31
PI-51
P1-62
P1-71
P2-16
P2-21
P2-46
P2-47
P2-48
P2-49
P2-50
P2-51
P2-52
P2-53
P2-54
P2-56
P2-61
P2-69

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order~ the contractor may use his own or any other facilities suitable for the performance of the inspection *requirements* specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. **A1l** items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system *or* quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated ***or*** actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified *as* follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see **4.4)** .

c. Special sampling inspection (see 4.5) .

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50° to +90°F.
b.	Relative humidity	90 percent RH max.
c.	Pressure	28 to 32 inches of Hg.

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
CMB Assemblies	9360878	3

These assemblies shall be manufactured in the same manner, using the same materials, equipment, processes and procedure as in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests, specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examinations **and tests**.

4.4.2.1 Components, assemblies and subassemblies.

a. **The** contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of **QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.**

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a Minimum the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Goverment for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) Provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table XX shall be performed on a 100 percent basis.

TABLE xx* Functional tests (100 percent inspection).

Characteristic	**Requirement**	**Vest Procedures**
Critical	None	N/A
Major **101.** Electrical 102. Performance	 3.4 3.5	 4.7.2 4.7.3
Minor	None	N/A

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4.5 Special sampling. one CMB out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. Each sample shall have met the requirements and tests in Table XX and shall then meet the requirements and tests in Table XXI . Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XXI. Special sampling.

Characteristic	Requireement	Test procedures
Critical	None	N/A
Major 103. Vibration 104. Shock 105. Altitude 106. Temperature 107. Humidity	 3.3.1 3.3.2 3.3.3 3.3.4 3.3.5	 4.7.1.1 4.7.1.2 4.7.1.3 4.7.1.4 4.7.1.5
Minor	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements~ acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test. as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment . Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment .

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in 'Examination" and "Test Facilities' requirements of MIL-F-13926 wherever Government

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designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table XXII (Test equipment list) shall be utilized in the performance of the applicable test methods and procedures as specified in 4.7. These items shall be assembled at the AN/USM-465A test station prior to test execution. Proper operation of interface devices and test equipment shall be verified before using same to run diagnostic tests.

TABLE XXII. Test equipment list.

Item	Number/Description	Drawing
1.	Interface Device TPS Adapter Board A	9361380
2.	Cable Assembly WI (Qty 2)	9397497
3.	Test Plug P3	9397498
4.	Test Plug P5	9397499 "
5*	Digital Card Tester	AN/USM-465A
6.	ComCode Program Tape	Version 2.0
7.	Test Program Set Manual	9361334
8.	Test Program Set Tape	TPT9361334

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

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4.7.1.1 **Vibration** **The test shall be performed as specified in Method 514, Category 8, Procedure I of MIL-STD 810 and 3.3.1 herein. In Step 1 of Procedure 1, only steps 4 through 8 of paragraph II-2.1 apply. The CMB shall be attached to the vibration generator directly or with a fixtures and securely held by its normal means of attachment. After vibration, the CMB shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.**

4.7.1.2 Shock. This test shall be performed as specified in Procedure 1, Section II of Method 516 of MIL-STD-810 except that:

a. **The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.**

b. **The shock pulses shall be 15 g's ± 10 percent amplitude and 11 ms ± 10 percent duration.**

c. **The maximum g shall occur at 5.5 milliseconds * 10 percent.**

d. **The shock measuring system shall have a flat frequency response within ± 10 percent from 20 to 2,000 hertz.**

After the application of all the shock impulses, the CMB shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 **Altitude** (operating) **This test shall be performed as specified in Procedures II, Method 500 of MIL-S-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.**

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) . This test shall be performed as specified in Procedures I and 11, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2, 4.7.3, for compliance with 3.3.4.

4.7.1.4.2 **Low (oerating and non-operating} . This test shall be performed as specified in Procedures I and II Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.**

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4.7.1.5 Humidity (non-operational **This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.**

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance The performance of the CMB shall be tested to meet the requiraents of 3.5 using the test equipment listed in Table XXII. The perfomnce testing shall be in accordance with the CMB Test Program Set Manual (TPS) drawing 9361334.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in 3.7.

5. **PACKAGING**

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360878. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The CMB assembly is used in the Fire Control Processor in the VADS self-propelled and towed vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. **Issue of** DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1) .

c. Requirements for submission of first article sample.

d. **Applicable stock number.**

e. Packaging requirements, if other than specified in Section 5.

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f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a. Inspection equipment designs for approv al Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

a. Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance Provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, DeveloPment and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development command (ARRADCOM), Fremkford Arsenal, Rock Island Arsenal or Picatlnny Arsenal drawings. Technical data originally prepared by these activities is now-under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Built in Test (BIT)
Fire Control Processor
Fire Control System
Vulcan Air Defense System (VmS)

6.6 Chances from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

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STANDARDIZATION OFFICE

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